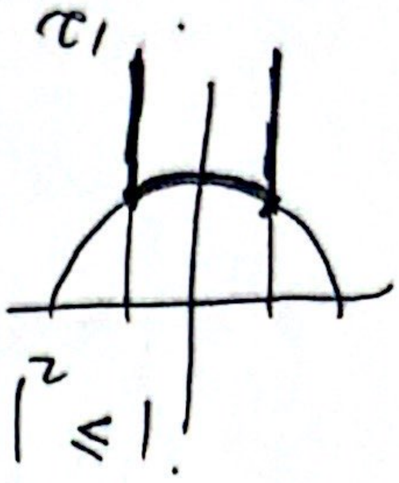
2.3.1 Suppose $\tau_1 \neq \tau_2 \in D$, $\tau_2 = \gamma\tau_1$, $\exists \gamma \in SL_2(\mathbb{Z})$

Then (1) $\begin{cases} \mathrm{Re}\,\tau_1 = \pm\frac{1}{2} \\ \tau_2 = \tau_1 \mp 1 \end{cases}$ or (2) $\begin{cases} |\tau_1| = 1 \\ \tau_2 = -\frac{1}{\tau_1} \end{cases}$.



Pf: Assume $\mathrm{Im}\,\tau_2 \geq \mathrm{Im}\,\tau_1$. let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Then $\mathrm{Im}\,\tau_2 = \dfrac{\mathrm{Im}\,\tau_1}{|c\tau_1+d|^2} \geq \mathrm{Im}\,\tau_1 \Rightarrow |c\tau_1+d|^2 \leq 1$.

$\tau_1 \in D \Rightarrow \mathrm{Im}\,\tau_1 \geq \frac{\sqrt{3}}{2}$.

Thus $\cancel{|c|\frac{\sqrt{3}}{2} \leq}$

$1 \geq |c\tau_1+d| \geq |\mathrm{Im}(c\tau_1+d)| = |c|\,\mathrm{Im}\,\tau_1 \geq |c|\frac{\sqrt{3}}{2}$.

$\Rightarrow |c| \leq \frac{2}{\sqrt{3}}$. $\Rightarrow |c| = 0, 1$.

① $c = 0$. $\gamma = \begin{pmatrix} \pm 1 & b \\ 0 & \pm 1 \end{pmatrix}$. Then $\mathrm{Re}\,\tau_2 = \mathrm{Re}\,\tau_1 + b$.

$\mathrm{Im}\,\tau_2 = \mathrm{Im}\,\tau_1$

Thus $|b| = 1$. and (1) holds.

② $|c| = 1$ Thus $|c\tau_1+d|^2 \leq 1 \Rightarrow |\tau_1+d|^2 \leq 1$.

$\Rightarrow (\mathrm{Re}\,\tau_1 \pm d)^2 + (\mathrm{Im}\,\tau_1)^2 \leq 1$.

$\Rightarrow (\mathrm{Re}\,\tau_1 \pm d)^2 \leq 1 - (\mathrm{Im}\,\tau_1)^2 \leq 1 - \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4}$

$\Rightarrow \mathrm{Re}\,\tau_1 \pm d \leq \frac{1}{2} \Rightarrow |d| = 0, 1$.

②-1 $|d| = 1$ $\mathrm{Re}\,\tau_1 = \pm\frac{1}{2}$. $\mathrm{Im}\,\tau_1 = \frac{\sqrt{3}}{2}$. and (1),(2) both hold

②-2 $d = 0$. $|c\tau_1+d| = |\tau_1| \leq 1$. $\Rightarrow |\tau_1| = 1$.

$\Rightarrow |\tau_2| = |\gamma\tau_1| = |\gamma| \cdot |\tau_1| = 1$.

and $\mathrm{Im}\,\tau_2 = |\tau_1|\,\mathrm{Im}\,\tau_1 = \mathrm{Im}\,\tau_1$.

Thus $\tau_2 = -\mathrm{Re}\,\tau_1 + i\,\mathrm{Im}\,\tau_1 = -\dfrac{|\tau_1|}{\tau_1} = -\dfrac{1}{\tau_1}$.

2.3.2 If $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ fixes $\tau \in \mathcal{H}$, show $|a+d| < 2$.

Pf: $\dfrac{a\tau+b}{c\tau+d} = \tau \Rightarrow a\tau+b = c\tau^2 + d\tau$.

$\Rightarrow c\tau^2 + (d-a)\tau \pm b = 0$ has two imaginary roots $\Rightarrow (d-a)^2 + 4bc < 0$.

$ad - bc = 1$. Thus $(d-a)^2 + 4ad - 4 < 0$

$(d+a)^2 < 4 \Rightarrow |a+d| < 2$.

2.3.3 If $\gamma \in SL_2(\mathbb{Z})$ has order 2, then $\gamma = -I$.

Pf: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 - bc & ab - bd \\ ac - cd & bc - d^2 \end{pmatrix} = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$.

$\Rightarrow \begin{cases} a^2 - bc = bc - d^2 = 1 \\ b(a-d) = c(a-d) = 0 \end{cases} \Rightarrow \begin{cases} a = \pm 1 \\ b = 0 \\ c = 0 \\ d = \pm 1 \end{cases} \Rightarrow \gamma = -I$.

2.3.4. (a) Show $L = \mathbb{Z}u + \mathbb{Z}v$ imply $\det(u\ v) = \pm 1$.

$L = \mathbb{Z}^2$.

Pf: This means $\forall x \in \mathbb{Z}^2\ \exists m, n \in \mathbb{Z}\ mu + nv = x$.

i.e. $(u\ v)\begin{pmatrix} m \\ n \end{pmatrix} = x$ has solution $\forall x$.

Let $x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, then $(u\ v)\begin{pmatrix} m_1 & m_2 \\ n_1 & n_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow \det(u\ v)\det\begin{pmatrix} m_1 & m_2 \\ n_1 & n_2 \end{pmatrix} = 1 \Rightarrow \det(u\ v) = \pm 1$.

(2.3.4) Prp 2.3.3 $\gamma \in SL_2(\mathbb{Z})$. (b) Prove Prp 2.3.3(a)(b).

(b) If $\operatorname{ord}\gamma = 4$ then γ conjugate to $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{\pm 1}$ in $SL_2(\mathbb{Z})$.

Pf: $\gamma^4 = I$. $\Rightarrow L^2$ is module over $\mathbb{Z}[i]$.

$$(a+bi)\cdot v := (aI + b\gamma)v.$$

$\mathbb{Z}[i]$ is PID. L is finitely generated over $\mathbb{Z}[i]$.

Thus $L \simeq \bigoplus_k \mathbb{Z}[i]/I_k$ as $\mathbb{Z}[i]$-modules.

Every $I_k \triangleleft \mathbb{Z}[i]$ has rank 2. as group.

L is free of rank 2 as abelian group.

Thus $\mathbb{Z}[i]/I_k$ is quotient group $\forall k$.

Thus $\nexists\, \phi_\gamma : \mathbb{Z}[i] \to L$. hence not exist $\mathbb{Z}[i]$-isomorphism.
-module

Let $u = \phi_\gamma(1)$, $v = \phi_\gamma(i)$, then $L = \mathbb{Z}u \oplus \mathbb{Z}v$.

$\det(u, v) = \pm 1$.

$$\gamma u = i \cdot \phi_\gamma(1) = \phi_\gamma(i) = v,$$
$$\gamma v = i \cdot \phi_\gamma(i) = -\phi_\gamma(1) = -u.$$

Thus $\gamma(u, v) = (v, -u) = (u\ v)\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

$$\Rightarrow \gamma = (u\ v)\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}(u\ v)^{-1}.$$

$$\gamma(v\ u) = (-u\ v) = (v\ u)\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{-1}. \qquad \Rightarrow \gamma = (v\ u)\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}(v\ u)^{-1}.$$

(a) If $\operatorname{ord}\gamma = 3$ then γ conjugate to $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}^{\pm 1}$ in $SL_2(\mathbb{Z})$.

Pf: $\gamma^3 = I \Rightarrow (-\gamma)^6 = I$.

Thus ~~Since~~ $-\gamma$ conjugate to $\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}^{\pm 1}$, γ conjugate to $-\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}^{\pm 1}$

$$-\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \qquad -\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}^{-1} = -\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}^{-1}.$$

2.3.6. i) In the proof of Cor 2.3.5, need the 2d points in $Y(\Gamma)$ listed in $E_\Gamma = \{\Gamma\gamma_j(i), \Gamma\gamma_j(\mu_3) : 1 \le j \le d\}$ be distinct?

$$SL_2(\mathbb{Z}) = \bigcup_{j=1}^{d} \Gamma\gamma_j.$$

Sol: This is asking whether $\exists \gamma \in \Gamma$ and $\exists j$

$$\Gamma\gamma_j(i) = \Gamma\gamma_j(\mu_3).$$

That is, if $\exists \gamma \in \Gamma$ $\gamma\gamma_j(i) = \gamma_j(\mu_3)$.

i.e. if $\exists \gamma \in SL_2(\mathbb{Z})$ $\gamma(i) = \mu_3$.

By Lem 2.3.2 this is impossible. Thus yes.

ii) Need the points of E_Γ all be elliptic points of Γ?

Sol: This is asking if $\gamma\gamma_j(i)$ or $\gamma\gamma_j(\mu_3) \in E_\Gamma$.

then $\exists \delta \in \Gamma$ $\delta\gamma\gamma_j(i) = \gamma\gamma_j(i)$ or $\delta\gamma\gamma_j(\mu_3) = \gamma\gamma_j(\mu_3)$

This is equivalent to if $\exists \delta, \gamma \in \Gamma$ $\gamma^{-1}\delta\gamma(\gamma_j(i)) = \gamma_j(i)$.

or $\gamma^{-1}\delta\gamma(\gamma_j(\mu_3)) = \gamma_j(\mu_3)$

Points in $SL_2(\mathbb{Z})i$ and $SL_2(\mathbb{Z})\mu_3$ may be not all elliptic points for Γ. Thus no.

2.3.7 Prove there is no elliptic point for the following groups

(a) $\Gamma(N)$, $N > 1$.

Pf: Since for elliptic point τ and $\gamma\tau = \tau$,

γ has order $1, 2, 3, 4$ or 6.

By Prp 2.3.3, $\operatorname{tr}\gamma = |a+d| = 0$ or ± 1. ~~γ could only have order 1 or 2~~,

which gives $\gamma = \pm I$. which means $\{\pm I\}$ trivial.

(b) $\Gamma_1(N)$ for $N > 3$. Also find elliptic points for $\Gamma_1(2)$ and $\Gamma_1(3)$ given that each group has one.

Pf: We need to show $\gamma \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} (N)$ $N > 3$ implies γ conjugate to $\pm I$.

The reason is same with (a), $\begin{cases} a+d = 2 \ (N) \\ |a+d| < 2. \end{cases}$

For $N = 3$ may have elliptic points since there could be

(e.g. $\begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix} \in \{\pm I\}$ $\Gamma_1(3)$.) $\Rightarrow a + d = -1$. Thus $x^2 + x + 1 = 0$ has 3 elliptic elements.

~~To see elliptic point, $\begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}\tau = -(2\tau - 1) = \tau$~~

conjugate to $S = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}^{\pm 1}$, $I = (ST)^2$. elliptic point $\mu_3 = e^{\frac{2\pi i}{3}}$

For $N = 2$, $a + d = 0$. $x^2 + 1 = 0$. has 2 elliptic elements

γ conjugate to $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{\pm 1}$, elliptic point i

(2.3.7)

(c) $\Gamma_0(N)$ for $p \mid N$ $p \equiv -1$ (12).

Pf: If $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$, then $c \equiv 0$ (p).

Thus $\gamma \equiv \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ $(p) \Rightarrow \begin{cases} d \equiv a^{-1} & (p). \\ a+d = 0 \text{ or } \pm 1 \end{cases}$

gives $a+d=0 \Rightarrow a^2 \equiv -1$ (p) $\Rightarrow$ -1 quadratic residue (p).

$a+d=\pm 1 \Rightarrow a^2 \mp a+1 \equiv 0$ (p) $\Rightarrow \Delta = -3$ quadratic residue (p).

$p \equiv -1$ (12) $\Rightarrow \begin{cases} p \equiv 3 & (4) \\ p \equiv 2 & (3). \end{cases}$

by Thm: -1 is quadratic residue mod p

$\Leftrightarrow p \equiv 1$ (4).

2.3.8 Suppose $SL_2(\mathbb{Z}) = \bigcup_j \{\pm I\}\Gamma\gamma_j$, Γ congruence subgroup of $SL_2(\mathbb{Z})$.

Show $\bigcup_j \gamma_j D \twoheadrightarrow Y(\Gamma)$.

Pf: $Y(\Gamma) = \{\Gamma\tau : \tau \in \mathcal{H}\}$. $\forall \tau \in \mathcal{H}$ $\exists \gamma \in SL_2(\mathbb{Z})$, $\tau' \in D$

~~$\tau' = \gamma\tau \in D$~~ $\tau = \gamma\tau'$. ~~$\Rightarrow \tau' = \gamma^{-1}\tau$~~

Moreover $\forall \gamma \in SL_2(\mathbb{Z})$ $\forall \delta \in \{\pm I\}\Gamma$ $\exists$ ~~and some~~ j ~~$= 1,$~~ $\gamma = \delta\gamma_j$.

This means $\forall \Gamma\tau \in Y(\Gamma)$ $\exists j$, $\tau' \in D$

~~$\gamma_j^{-1}\delta^{-1}\tau$~~ $\delta\gamma_j\tau' = \tau$.

i.e. $\gamma_j\tau' = \delta^{-1}\tau$ $\Rightarrow \bigcup_j \gamma_j D \twoheadrightarrow Y(\Gamma)$

$\forall \delta \in \Gamma$